

plants have inherited two purple-flower alleles; clearly, these plants will have purple flowers. One-half of the F_2 offspring have inherited one purple-flower allele and one white-flower allele; these plants will also have purple flowers, the dominant trait. Finally, one-fourth of the F_2 plants have inherited two white-flower alleles and will express the recessive trait. Thus, Mendel's model accounts for the 3:1 ratio of traits that he observed in the F_2 generation.

Useful Genetic Vocabulary

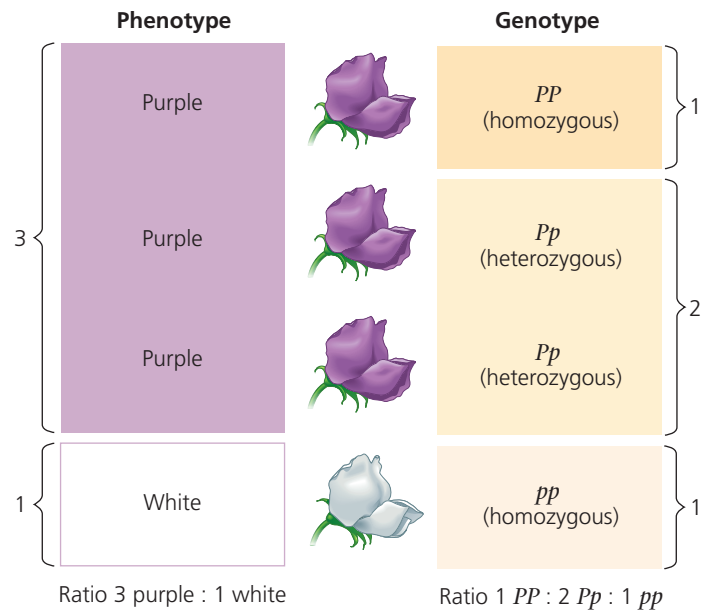
An organism that has a pair of identical alleles for a gene encoding a character is called a **homozygote** and is said to be **homozygous** for that gene. In the parental generation in Figure 14.5, the purple-flowered pea plant is homozygous for the dominant allele (PP), while the white plant is homozygous for the recessive allele (pp). Homozygous plants “breed true” because all of their gametes contain the same allele—either P or p in this example. If we cross dominant homozygotes with recessive homozygotes, every offspring will have two different alleles— Pp in the case of the F_1 hybrids of our flower-color experiment (see Figure 14.5). An organism that has two different alleles for a gene is called a **heterozygote** and is said to be **heterozygous** for that gene. Unlike homozygotes, heterozygotes produce gametes with different alleles, so they are not true-breeding. For example, P - and p -containing gametes are both produced by our F_1 hybrids. Self-pollination of the F_1 hybrids thus produces both purple-flowered and white-flowered offspring.

Because of the different effects of dominant and recessive alleles, an organism's traits do not always reveal its genetic composition. Therefore, we distinguish between an organism's appearance or observable traits, called its **phenotype**, and its genetic makeup, its **genotype**. As shown in Figure 14.5 for the case of flower color in pea plants, PP and Pp plants have the same phenotype (purple flowers) but different genotypes. **Figure 14.6** reviews these terms. Note that the term *phenotype* refers not only to traits that relate directly to physical appearance but also to physiological traits. For example, one pea variety lacks the normal ability to self-pollinate, which is a phenotypic trait (called non-self-pollination).

The Testcross

Given a purple-flowered pea plant, we cannot tell if it is homozygous (PP) or heterozygous (Pp) because both genotypes result in the same purple phenotype. To determine the genotype, we can cross this plant with a white-flowered plant (pp), which will make only gametes with the recessive allele (p). The allele in the gamete contributed by the purple-flowered plant of unknown genotype will therefore determine the appearance of the offspring (**Figure 14.7**). If all the offspring of the cross have purple flowers, then the purple-flowered mystery plant must be homozygous for the dominant allele, because a $PP \times pp$ cross produces all Pp offspring. But if both the purple and the white phenotypes

▼ **Figure 14.6 Phenotype versus genotype.** Grouping F_2 offspring from a cross for flower color according to phenotype results in the typical 3:1 phenotypic ratio. In terms of genotype, however, there are actually two categories of purple-flowered plants, PP (homozygous) and Pp (heterozygous), giving a 1:2:1 genotypic ratio.



appear among the offspring, then the purple-flowered parent must be heterozygous. The offspring of a $Pp \times pp$ cross will be expected to have a 1:1 phenotypic ratio. Breeding an organism of unknown genotype with a recessive homozygote is called a **testcross** because it can reveal the genotype of that organism. The testcross was devised by Mendel and continues to be used by geneticists.

The Law of Independent Assortment

Mendel derived the law of segregation from experiments in which he followed only a *single* character, such as flower color. All the F_1 progeny produced in his crosses of true-breeding parents were **monohybrids**, meaning that they were heterozygous for the one particular character being followed in the cross. We refer to a cross between such heterozygotes as a **monohybrid cross**.

Mendel worked out the second law of inheritance by following *two* characters at the same time, such as seed color and seed shape. Seeds (peas) may be either yellow or green. They also may be either round (smooth) or wrinkled. From single-character crosses, Mendel knew that the allele for yellow seeds is dominant (Y), and the allele for green seeds is recessive (y). For the seed-shape character, the allele for round is dominant (R), and the allele for wrinkled is recessive (r).

Imagine crossing two true-breeding pea varieties that differ in *both* of these characters—a cross between a plant with yellow round seeds ($YYRR$) and a plant with green wrinkled seeds ($yyrr$). The F_1 plants will be **dihybrids**, individuals heterozygous for the two characters being followed in the cross ($YyRr$). But are these two characters transmitted from parents to